

Experiment No 1

Determine the Amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
In 500cm^3 of Given Sample Solution You
Are Provided With 0.025M EDTA:

Principle:-

Complex-metric situation

Standard Solution:-

0.025M EDTA

Indicators :-

Esichsome Black T

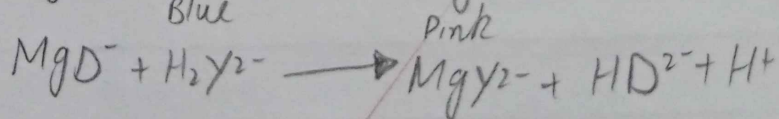
End Point :

Appearance of blue colour

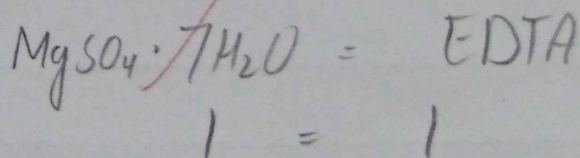
Buffer Solution:

$$\text{NH}_3/\text{NH}_4\text{Cl} \neq 2.0 \text{ cm}^3 \quad / \text{pH} = 10$$

Chemical Reaction:



Mole Ratio:-

Observations

Experiment No: 1

Determine The Amount Of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

In 500cm^3 Of Given Sample

Solution You Are Provided With

0.025M EDTA: (372)

Principle:

It is a complexometric titration amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ is determined by calculating the molarity $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. EDTA is used as a standard solution. Buffer solution is used to maintain the pH of solution. **Erichrome Black T** is used as an indicator which show wine red colour.

Purpose

The purpose of indicator is to set free metal ions which form complex with EDTA.

End Point

The end point is the appearance of blue colour

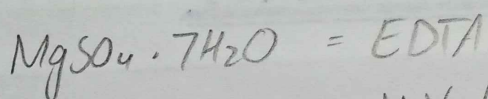
Procedure:

Take 0.025M EDTA in burette

Sr. No	Initial (cm ³) Reading	Final (cm ³) Reading	Volume (cm ³) used.
1-	0.0	12.8	12.8 cm ³
2-	13.0	25.8	12.8 cm ³
3-	26.0	38.8	12.8 cm ³

Mean Volume = 12.8 cm³

Calculation:-



$$M_1 V_1 / n_1 = M_2 V_2 / n_2$$

$$M_1 = 0.025 \times \frac{12.8}{10}$$

$$M_1 = 0.032 \text{ M}$$

Molarity of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O} = 0.032 \text{ M}$

Amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 1000 cm³ of EDTA = Molarity $\times$ At. wt

$$= 0.032 \times 246$$

$$= 7.872 \text{ g/dm}^3$$

Amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 500 cm³ of EDTA

$$= 7.872 / 2$$

$$= 3.936 \text{ g/dm}^3$$

Result:-

Amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 500 cm³ of given sample solution is 3.936 g/dm³

⑧ Pipette out 10cm^3 of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in titration flask.

⑧ Add 2ml of buffer solution in titration flask shake it well.

⑧ Then add four drops of Eriochrome Black T as an indicator in titration flask.

⑧ The wine red colour appear.

⑧ Titrate it with standard solution of EDTA till blue colour appeared.

⑧ It is end point. Take three concordant reading in similar way.

Result:-

Amount of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 500cm^3 of given sample solution is **3.936g** .

Experiment NO 2:

Determine the Amount of $ZnSO_4 \cdot 7H_2O$ In
 dm^3 of Given Sample Solution and Find
the Amount of Zn^{2+} Ions/ dm^3 You Are
Prepared with 0.025M EDTA Solution:

Principle: Complexometric titration

Standard Solution:

0.025M EDTA

Indicator:

Erichrome Black T

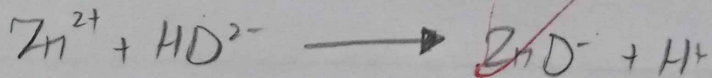
End point:

Appearance of blue colour

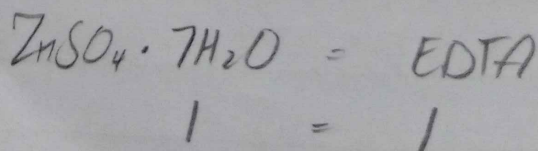
Buffer Solution:-

$NH_3/NH_4Cl = 2.0 cm^3$ / $pH = 10$

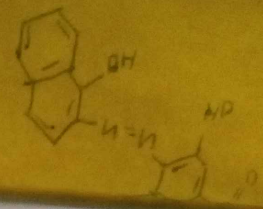
Chemical Reaction:



Mole Ratio:



Experiment No: 2



Determine The Amount Of $ZnSO_4 \cdot 7H_2O$
In dm^3 Of Given Sample solution Al-
so Find Out The Amount Of Zn^{2+} Ions
/ dm^3 . You Are Provided with 0.025M EDTA

Solution:-

Principle:-

It is Complexometric titration Amount of $ZnSO_4 \cdot 7H_2O$ (amount) and Zn^{2+} can be determined by calculating the molarity of $ZnSO_4 \cdot 7H_2O$

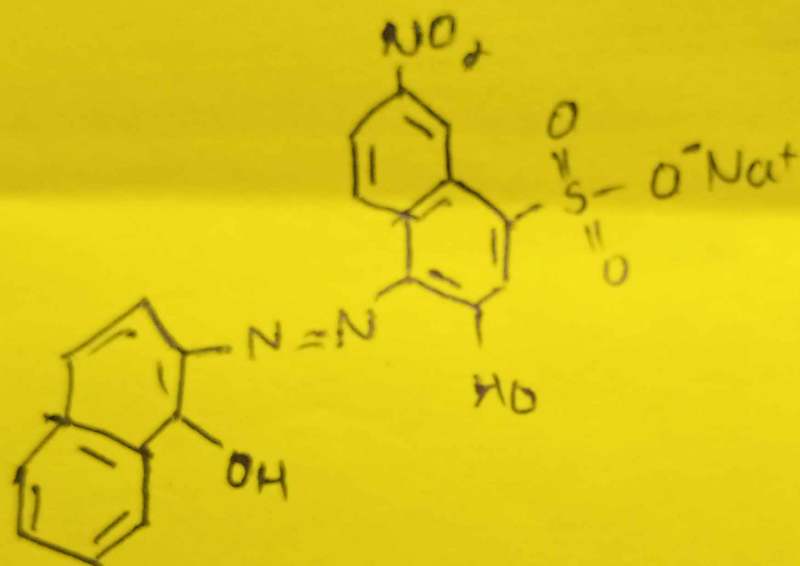
Standard Solution

EDTA is used as standard solution and its molarity is 0.025M

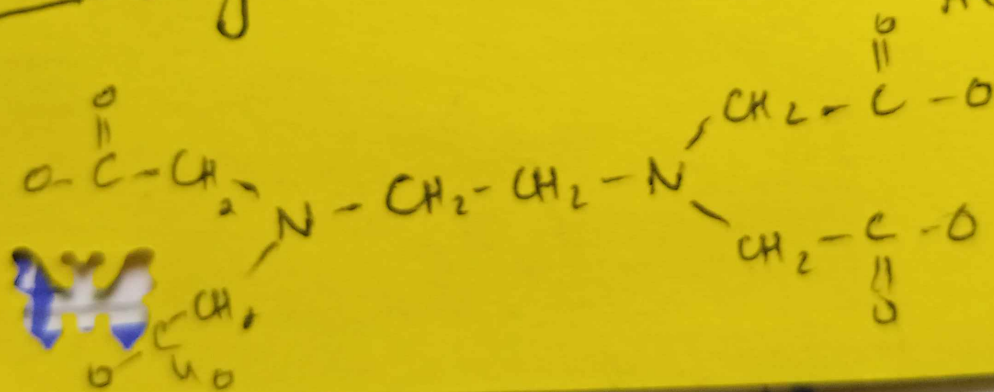
Buffer Solution

Buffer solution is used to maintain the pH of the solution at 10.

Erichrome Black T₈



EDTA: Ethylene Diamine Tetra Acetic Acid



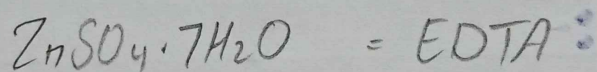
0.025M EDTA

Observation:

Sr. NO	Initial Reading (cm ³)	Final Reading (cm ³)	Volume Used (cm ³)
1-	0.0	13.8	12.8 cm ³
2-	13.0	26.8	12.8 cm ³
3-	26.0	38.8	12.8 cm ³

Mean Volume of EDTA used = 12.8 cm³

Calculation:



$$M_1 V_1 / n_1 = M_2 V_2 / n_2$$

$$M_1 = \frac{0.025 \times 12.8}{10.0}$$

$$M_1 = 0.032 \text{ M}$$

$$\text{Molarity of } \text{ZnSO}_4 \cdot 7\text{H}_2\text{O} = 0.032 \text{ M}$$

$$\begin{aligned} \text{Amount of } \text{ZnSO}_4 \cdot 7\text{H}_2\text{O} &= \text{Molarity} \times \text{Atomic wt.} \\ \text{in } 1 \text{ dm}^3 &= 0.032 \times 287.4 \end{aligned}$$

$$= 9.1968 \text{ g/dm}^3$$

$$\begin{aligned} \text{Amount of } \text{Zn}^{2+} \text{ ions in } 1 \text{ dm}^3 &= \text{Molarity} \times \text{At. wt.} \\ &= 0.032 \times 65.39 \end{aligned}$$

$$= 2.076 \text{ g/dm}^3$$

Indicator

Erichrome Black T is used as an indicator. It shows wine red colour. The purpose of the indicator is to set free the metal ions. These free metal ions form Coordination Complex with EDTA.

End Point

The end point is appearance of the blue colour.

Procedure:-

☞ Take 0.025M EDTA in burette.

☞ Pipette out 10.0 cm^3 of $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ in the titration flask.

☞ Take 2.0 cm^3 of the buffer solution and added in the titration flask. Shake it very well.

☞ Then add four drops Erichrome Black T in titration flask. Solution become wine red in colour.

☞ Then titrate against the EDTA till blue colour appeared. It is end point

☞ Repeat the experiment to get three concordant readings.

Result:-

⇒ Amount of $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ in given sample solution = 9.1968 g/dm^3

⇒ Amount of Zn^{2+} in given sample solution is = 2.076 g/dm^3

il

Also take blank reading by using solvent (distilled water).

Take 10.0cm^3 of distilled water add 2.0cm^3 of buffer solution and four drops of indicator.

Then titrate it against EDTA and note reading. Subtract this reading from volume of EDTA used. This is correct reading.

Result:-

The amount of $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ in given sample solution is 9.1968g/dm^3

Amount of Zn^{2+} in given sample solution is 2.076g/dm^3

Experiment No. 3:

Determine the Amount of $\text{NiSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 500cm^3 of given Sample Solution using 0.01M EDTA and 0.01M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ By Back Titration Method.

Principle: Complexometric Back titration

Standard Solution:

0.01M EDTA = 0.01M MgSO_4

Indicators:

Erichrome Black T

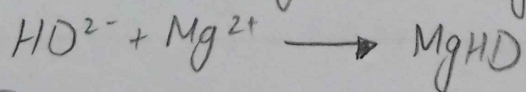
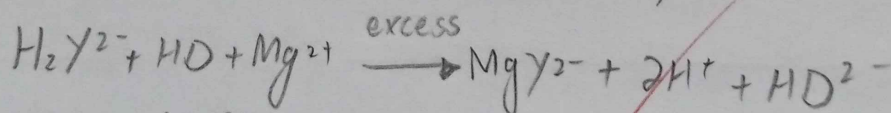
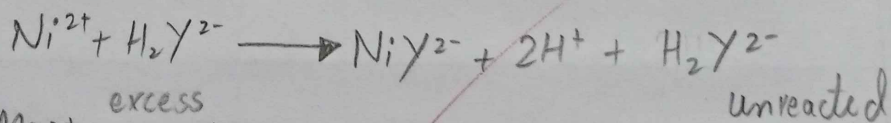
End point:

Blue to Wine red colour

Buffer Solution:

$\text{NH}_3/\text{NH}_4\text{Cl} = 2.0\text{cm}^3$ - $\text{pH} = 10$

Chemical Reaction:



Mole Ratio:-

$\text{NiSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O} = \text{EDTA}$

Observation:-

Sr NO	Initial volume cm^3	Final volume (cm^3)	Volume used (cm^3)
1-	0.0	12.0	12.0 cm^3
2-	12.0	24.0	12.0 cm^3
3-	24.0	36.0	12.0 cm^3

Experiment No. 3:

Determine the Amount of NiSO_4
 $(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 500cm^3 of Given Sam-
ple Solution Using 0.01M EDTA And
 0.01M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ By Back Titration

Method.

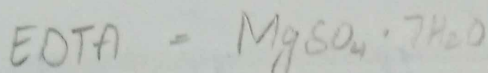
Principle:

This is an Complexometric back titration in which standard solution of EDTA and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ used that is complex forming agent. The other is 0.01M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.

In this titration we use excess amount of EDTA which form the complex with Ni^{+2} then we use $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. This $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ react with remaining EDTA. When all the EDTA is consumed then original colour of Eriochrome Black T (indicator) was reappeared.

From this we can measure the amount of EDTA consumed. EDTA is a strong complex forming agent. So it form strong complex with Ni^{+2} and shows blue colour.

Calculation:-



$$M_1 V_1 / n_1 = M_2 V_2 / n_2$$

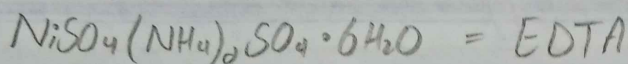
$$0.01 \times V_1 / 1 = 0.01 \times 12.0 / 1$$

$$V_1 = 12.0 \text{ cm}^3$$

→ Volume of EDTA used by $\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$

$$= 20.0 - 12.0$$

$$= 8.0 \text{ cm}^3$$



$$M_1 V_1 / n_1 = M_2 V_2 / n_2$$

$$M_1 = \frac{0.01}{10} \times 8.0 = 0.008 \text{ M}$$

Molarity of $\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is = 0.008 M

Amount of $\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 1 dm³ = Mol × Mol wt

$$= 0.08 \times 395$$

$$= 3.16 \text{ g/dm}^3$$

Amount of $\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 500 cm³ = 3.16 g/2

$$= 1.58 \text{ g}$$

Result:-

Amount of $\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 500 cm³ of given sample solution is 1.58 g.

But after titrating with $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ excess EDTA is consumed and colour was changed in the brick red.

Procedure:

☞ Take 0.01M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in burette.

☞ Pipette out 10.0cm^3 of the $\text{NiSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in titration flask.

☞ Add 2.0cm^3 of buffer solution and 4 drops of Eriochrome Black T indicator in the titration flask and shake it well.

☞ Add 20.00cm^3 of EDTA in titration flask. Shake it vigorously. Blue colour appeared.

☞ Now titrate it against the $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ till the wine red colour appeared.

☞ It is end point.

☞ Repeat the experiment to get the concordant readings.

Result:

The amount of $\text{NiSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in 500cm^3 of given sample solution is 1.58g.

Experiment No 4.

The given Solution Contain Mixture Of
 $ZnSO_4$ And $CaSO_4$. Determine Amount Of
 Ca^{2+} And Zn^{2+} Ions/dm³ By Masking Method
You Are Provided With 0.01M $MgSO_4 \cdot 7H_2O$ And
0.01M EDTA Solution.

Principle:

Complexometric back titration

Standard Solution:

0.01M $MgSO_4 \cdot 7H_2O$

0.01M EDTA solution

Indicator:-

Exochrome Black T

End Point:-

Appearance of wine red colour.

Chemical Reaction:-



Experiment No. 4:

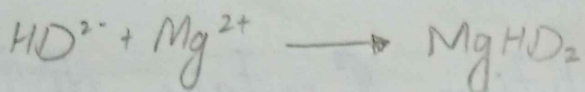
The Given Solution Contain Mixture
Of ZnSO_4 And CaSO_4 . Determine Amo-
unt Of Ca^{2+} And Zn^{2+} Ions/ dm^3 By
Masking Method You Are Provided With
0.01M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ And 0.01M EDTA So-
lution.

Principle :-

This is Complexometric back titration in which two standard solution $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ and 0.01M EDTA is used.

Determine the amount of Ca^{2+} and Zn^{2+} ions by calculating the molarity of the Zn^{2+} and Ca^{2+} ions. Excess amount of EDTA is used in first step. But in the second step KCN is used which mask Zn^{2+} ion and form stable complex.

$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ molecules form the complex with Ca^{2+} ions. So by masking the one agent the molarity of second agent is determined.



Mole Ratio :-

Mixture = EDTA

1 = 1

Titration I :-

(Observation)

Sr. No.	Initial Volume (cm ³)	Final Volume (cm ³)	Volume used (cm ³)
1	0.0	2.0	2.0 cm ³
2	2.0	4.0	2.0 cm ³
3	4.0	6.0	2.0 cm ³

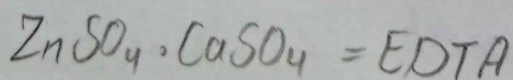
Mean Volume = 2.0 cm³

Calculation :-

Volume of EDTA used by mixture of

$$\text{ZnSO}_4 \cdot \text{CaSO}_4 = 40.0 - 2.0$$

$$= 38.0 \text{ ml}$$



$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2}$$

$$\frac{M_1 \times 10}{1} = \frac{0.01 \times 38.0}{10} \Rightarrow M_1 = 0.038 \text{ M}$$

Procedure:-

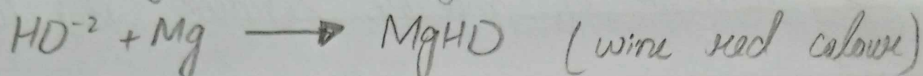
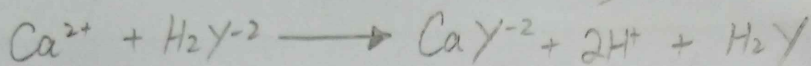
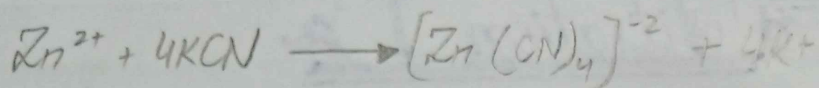
Titration I

- I took $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in burette.
- Take 10.0cm^3 of mixture solution in titration flask.
- Add 40ml of EDTA in same titration flask.
- Add 20cm^3 of buffer solution and add 2-3 drops of indicator erichrome Black T. Shake it well and blue colour appeared.
- Titrate it against $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ till wine red colour appeared.
- It is end point
- Repeat the experiment to get three concordant readings

Titration II

- In 2nd step take 10.0cm^3 of the mixture in the titration flask.
- Add 25ml of EDTA and 15ml of KCN carefully in the titration flask.
- Shake it well and add 2.0cm^3 of buffer solution.

Titration II :- of Reaction B



Observation:

Sr. No	Initial Reading (cm ³)	Final Reading (cm ³)	Volume used cm ³
1.	0.0 cm ³	1.5 cm ³	1.5 cm ³
2.	1.5 cm ³	3.0 cm ³	1.5 cm ³
3.	3.0 cm ³	4.5 cm ³	1.5 cm ³

Mean Volume Used = 1.5 cm³

Calculation:

EDTA used for Ca²⁺ ions = 25 - 1.5 = 23.5 ml

Ca²⁺ ions = EDTA

$$\frac{M_1 V_1}{n_1} = \frac{M_2 V_2}{n_2} \Rightarrow \frac{M_1 \times 10}{1} = \frac{0.01 \times 23.5}{1} \Rightarrow \boxed{M_1 = 0.0235\text{M}}$$

Molarity of Zn²⁺ ion = 0.038 - 0.0235 = $\boxed{0.0145\text{M}}$

Amount of Ca²⁺ ion per/dm³ = Mol x at. wt = 0.0235 x 40 = $\boxed{0.94\text{g/dm}^3}$

Amount of Zn²⁺ ion per/dm³ = Mol x At. wt = 0.0145 x 65.39 = $\boxed{0.95\text{g/dm}^3}$

Result:-

Amount of Ca²⁺ ions = 0.94 g/dm³

Amount of Zn²⁺ ions = 0.95 g/dm³

Also add few drops of the Eriochrome Black T indicator in titration flask.

KCN is used to mask Zn^{2+} ions.

Now titrate it against the $MgSO_4 \cdot 7H_2O$ till the wine red colour appear. It is end point.

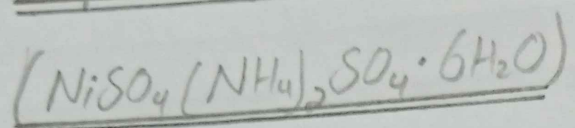
Repeat the experiment to get three concordant readings.

Result :-

The amount of Calcium ions in mixture solution is 0.94 g/dm^3

The amount of Zn^{2+} ions in mixture solution is 0.95 g/dm^3

Preparation

Experiment No 5:Preparation Of Nickel Ammonium Sulphate.Material Required:-

$$\text{NiSO}_4 \cdot 6\text{H}_2\text{O} = 7 \text{ grams}$$

$$(\text{NH}_4)_2\text{SO}_4 = 3.5 \text{ grams}$$

$$\text{Dilute H}_2\text{SO}_4 = 1 \text{ ml}$$

Reaction:-Calculation:-

From balance chemical equation

$$395 \text{ g of reactant} = 395 \text{ g of product}$$

$$1 \text{ gram of reactant} = \frac{395}{395} \times 1$$

$$\boxed{= 1}$$

Experiment No 5:

10

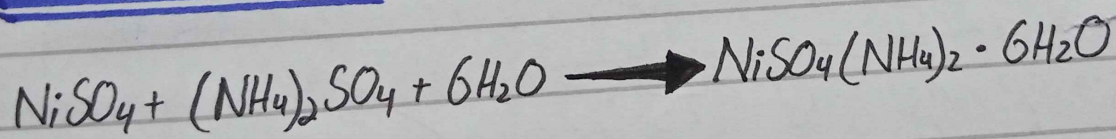
Preparation Of Nickel Ammonium Sulphate $(\text{NiSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O})$

Principle:-

When solution of nickel sulphate ammonium sulphate are mixed together and solution is concentrated by heating. And then cooled it and allow it to stabilize.

Pale green crystals of Nickel Ammonium sulphate are obtained.

Reaction:-



Procedure:-

☞ Took beakers and washed them with distilled water.

☞ Then dissolved 7 grams of NiSO_4 into 10 ml of distilled water containing 1 ml H_2SO_4 .

☞ Then weighed out 3.5 grams of Ammonium sulphate in 10 ml of distilled water.

10.5g of reactant = $\frac{395}{395} \times 10.5$

= 10.5g

% Yield:-

Theoretical yield = 10.5g

Actual yield = 7.9g

%age yield = $\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$

= $\frac{7.9}{10.5} \times 100$

%age yield = 75.2%

Result:-

The percentage yield of Nickel Ammonium Sulphate $[\text{NiSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}]$ is 75.2%

in another beaker.

☞ Mix both the solution of nickel Sulphate and ammonium sulphate in a beaker and mix them well.

☞ Now heat the above solution on H_2O bath and let it to evaporate and concentrate to half of its volume.

☞ Cool the above concentrated solution and allow it to crystallized and let it to stand for at least about one hour.

☞ Pale green crystals are formed in the solution.

☞ Now filter the crystals by using filter paper.

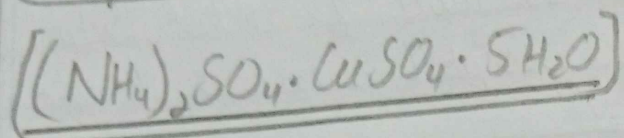
☞ After filtering the crystals wash the crystals with H_2O and then with alcohol and again with water.

☞ Then dry the crystals between the folds of filter paper.

☞ After drying the crystals weight them and find their yield.

Result:-

Percentage yield of Nickel Ammonium sulphate $[NiSO_4 (NH_4)_2 \cdot SO_4 \cdot 6H_2O]$ is 7.52.

Experiment No 6:Preparation of Ammonical Copper SulphateMaterial Required:-

Ammonium Sulphate = 2.0g

Copper Sulphate = 4.0g

Reaction:Calculations

381.5g of reactant = 381.5g of product

1.0g of reactant = $\frac{381.5}{381.5}$

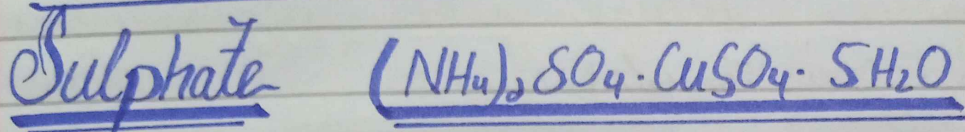
6 gram of reactant = $\frac{381.5}{381.5} \times 6$

= 6 gram

Experiment No 6:

19

Preparation Of Ammonical Copper Sulphate



Principle :-

When solution of Copper sulphate and ammonium sulphate are mixed together (double Sulphate) first heat and then cool them.

Crystals

Dry crystals of ammonium copper sulphate are obtained.

Reaction:-



Procedure :-

First of all wash the apparatus with distilled water.

Then dissolved 2g of Ammonium Sulphate in 10ml of distilled water.

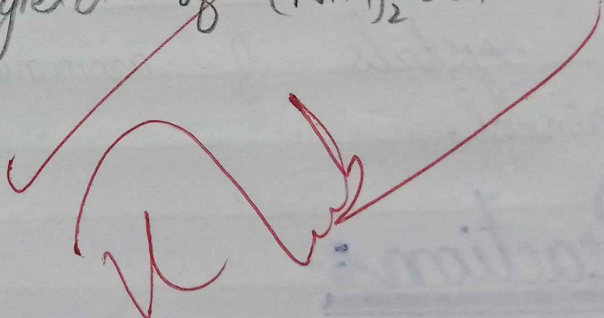
$$\% \text{ age yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

$$\% \text{ age yield} = \frac{3.8}{6} \times 100$$

$$\% \text{ age yield} = 63.3\%$$

Result:

Percentage yield of $(\text{NH}_4)_2\text{SO}_4 \cdot \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 63.3%.



Now measure 4g of copper Sulphate and dissolved it in 10ml of distilled H_2O

Mix both the solution and concentrate it about half of its volume.

Let it leave for three hours at room temperature for stabilization.

ppt's are formed and filtered them.

The ppt's of $(NH_4)_2CuSO_4 \cdot 5H_2O$ get on filter paper and dried them between the folds of filter paper.

Measure the percentage yield of $(NH_4)_2SO_4 \cdot CuSO_4 \cdot 5H_2O$

Result:-

The percentage yield of $(NH_4)_2SO_4 \cdot CuSO_4 \cdot 5H_2O$ is **63.3%**.

Experiment No 7:

5

Preparation Of Potassium Trioxalato Aluminate (III) - $K_3[Al(C_2O_4)_3]$

Material Required:

$$Al_2(SO_4)_3 = 2.1g / 15cm^3$$

$$NaOH = 2.00g / 15cm^3$$

$$K_2C_2O_4 \cdot H_2O = 2.7g$$

$$H_2C_2O_4 \cdot H_2O = 2.0g$$

Yield:-

✓
1.0 gram

Experiment NO 7:

6

Preparation Of Potassium Trioxalato Aluminate (III). $K_3[Al(C_2O_4)_3]$.

Principle:-

When solution of $Al_2(SO_4)_3$ and NaOH mixed together yellowish ppt of $Al(OH)_3$ are formed. These ppts are mixed with salt of potassium oxalate in the presence of oxalic acid, boil and filter the ppts and crystal of the potassium trioxalato aluminate $K_3[Al(C_2O_4)_3]$ are formed.

Procedure:-

- ☞ Wash the apparatus properly water.
- ☞ Now weighing all required.
- ☞ Now prepared the solution of $Al_2(SO_4)_3$ in 15ml of H_2O .
- ☞ Mixed both the solution to get the precipitates of $Al(OH)_3$. But on mixing the solution is turned yellow. It's mean that it is impure.

Result:-

1.0g of potassium trioxalato
Aluminate $K_3[Al(C_2O_4)_3]$ is obtained.

Done

So at that point, added excess NaOH so that the precipitates dissolved in it.

Now filter the solution and added 4-5 drops of HCl in it to re-precipitate the ppt. the ppts of $Al(OH)_3$.

Adjust its pH

Now add 40.0ml of solution of 2.76g of $K_2C_2O_4 \cdot 2H_2O + H_2C_2O_4 \cdot 2H_2O$ in the ppts of $Al(OH)_3$.

Now concentrated it by heating it up to its half volume.

Leave the above concentrated solution at room temperature and allow it to stand.

Filter out the crystals of potassium trioxalato Aluminate $K_3[Al(C_2O_4)_3]$

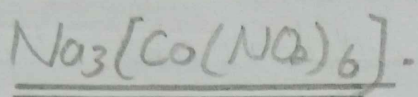
Calculate the percentage yield of $K_3[Al(C_2O_4)_3]$.

Result:-

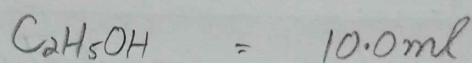
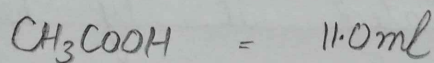
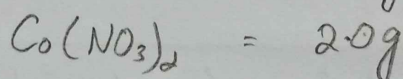
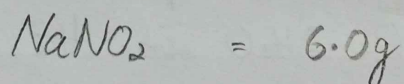
1.0g of potassium-trioxalato Aluminate $K_3[Al(C_2O_4)_3]$ is obtained.

Experiment No 8:

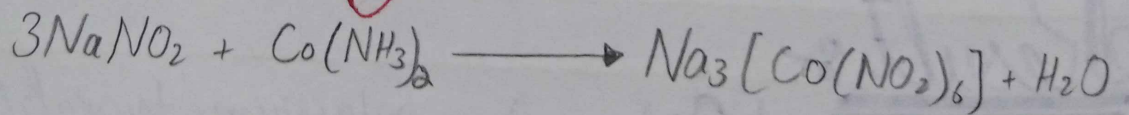
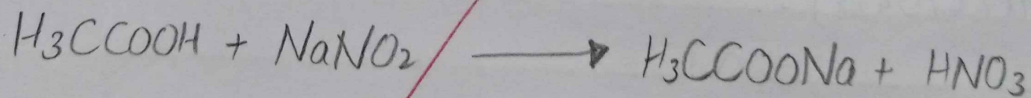
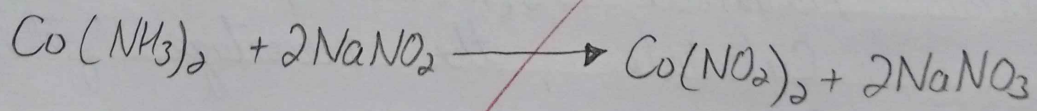
Preparation Of Sodium Cobalt Nitrate



Material Required:



Chemical Reaction:



Experiment No. 8:

2

Preparation Of Sodium Cobalt Nitrate $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$

Principle:-

Solution of NaNO_2 and $\text{Co}(\text{NO}_3)_2$ are mixed and then acidic acid added gradually in the mixture. Then ethyl alcohol is added in it to clear the solution.

Let it to be stable Overnight and ppts of sodium cobalt nitrate obtained after this filter them and dry them in open air.

Procedure:-

⌘ weighed 6.0g of NaNO_2 and dissolved it in 10ml of warm water.

⌘ Took 5g of $\text{Co}(\text{NO}_3)_2$ added by continuous stirring until it is completely dissolved.

⌘ Then added CH_3COOH gradually in this solution by constant stirring.

⌘ Filter to remove impurities and added 10ml of alcohol ($\text{C}_2\text{H}_5\text{OH}$) in small quantity to clear solution.

Result:-

Yield of sodium cobalt nitrate
 $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ is $\checkmark 1.8\text{g}$

ilub

Procedure:-

Left this solution overnight so that precipitation take place.

Now filter it and washed ppt with small quantity of alcohol until washing is colourless.

Allow it to dry in air and weight them.

Result :-

The yield of sodium cobalt nitrate $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ is 1.5g.